

# Kairix AI

## AI-Powered Career Coaching System

Vision AI · Speech AI · RAG · Automation

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AI Product Manager — Cohort 28 · May 2026



#### Vision AI

Google Cloud Vision API



#### Speech AI

Web Speech API + TTS



#### RAG System

LangChain + ChromaDB



#### Automation

n8n → Gmail + Sheets

# Problem & Solution

## THE PROBLEM

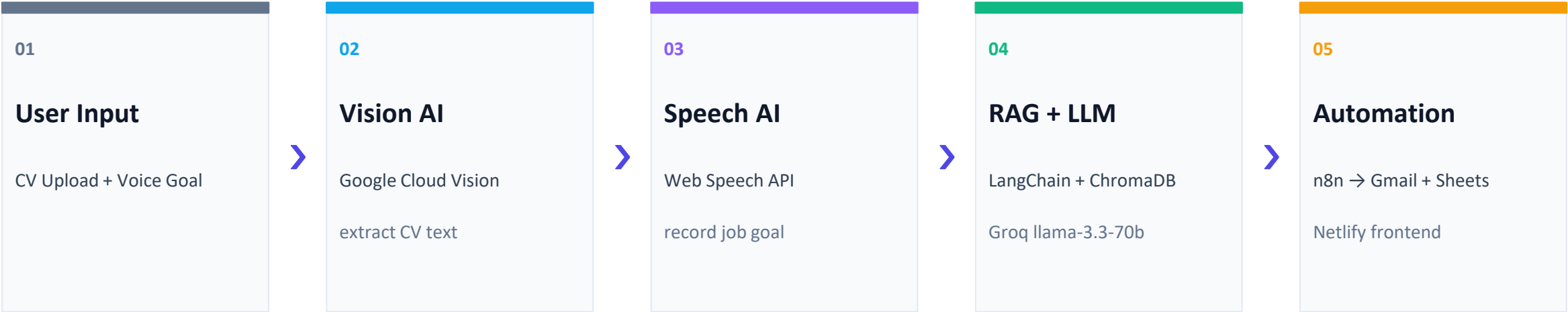
- Job seekers get rejected without knowing why
- Generic CV advice doesn't match real job market requirements
- Career coaching is expensive and slow — feedback takes days
- No affordable tool gives instant, role-specific, scored feedback



## THE SOLUTION — KAIRIX AI

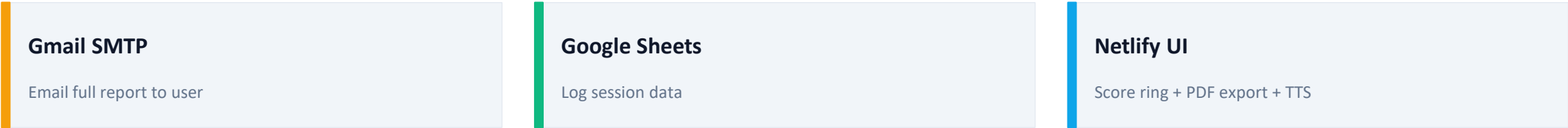
- Upload CV as JPG, PNG, PDF, or DOCX
- Speak your target role via microphone
- Receive a scored coaching report in under 10 seconds
- Full report emailed automatically via n8n + Gmail
- Session logged to Google Sheets for tracking

# System Architecture



*End-to-end pipeline: user input → AI processing → intelligent output → real-world automation*

**n8n Webhook splits into:**



# Four AI Technologies — One Unified System



## Vision AI

Google Cloud Vision API

- Analyses uploaded CV images using OCR
- Extracts all text from JPG, PNG, WEBP files
- pdfplumber handles PDF text extraction
- python-docx handles DOCX files



## Speech AI

Web Speech API + Web Speech Synthesis

- SpeechRecognition converts voice to text in real time
- User speaks their target job role — no typing needed
- Web Speech Synthesis reads back the full report aloud
- Works natively in Chrome and Edge — no library needed



## RAG System

LangChain + ChromaDB + Sentence Transformers

- Custom dataset: 2026 job market data & salary benchmarks
- all-MiniLM-L6-v2 embeds documents into ChromaDB
- Relevant context retrieved at query time via LangChain
- Injected into LLM prompt for role-specific accuracy



## Automation

n8n Webhook → Gmail SMTP + Google Sheets

- n8n webhook triggered after every analysis
- Full coaching report emailed to user automatically
- Session logged: timestamp, email, job goal, score
- Zero manual intervention — fully event-driven

# Tools & Technologies

|                      |                                    |                  |                                          |
|----------------------|------------------------------------|------------------|------------------------------------------|
| Frontend             | HTML5, CSS3, Vanilla JavaScript    | Embeddings       | Sentence Transformers — all-MiniLM-L6-v2 |
| Voice Input          | Web Speech API — SpeechRecognition | Vector Store     | ChromaDB                                 |
| Text-to-Speech       | Web Speech Synthesis API           | RAG Framework    | LangChain                                |
| Backend              | FastAPI — Python 3.10+             | AI Model         | Groq — llama-3.3-70b-versatile           |
| CV OCR (Images)      | Google Cloud Vision API            | Automation       | n8n → Gmail SMTP + Google Sheets         |
| CV Extraction (PDF)  | pdfplumber                         | Frontend Hosting | Netlify                                  |
| CV Extraction (DOCX) | python-docx                        | Backend Hosting  | Railway (nixpacks)                       |

# Live Demo & Repository

## LIVE DEMO

# kairix.netlify.app

Fully deployed — no login required · works in Chrome / Edge

*Analysis completes in under 10 seconds*

## GITHUB REPOSITORY

# github.com/MuhammadTahir-1-9-4/careerlens-ai

Full source code · FastAPI backend · HTML/CSS/JS frontend

*README with setup, architecture, and API reference*

## DEMO WALKTHROUGH

- 1 Open kairix.netlify.app in Chrome
- 2 Upload a CV (JPG / PDF / DOCX)
- 3 Click microphone → speak target role
- 4 Click 'Analyze My Career'
- 5 View score ring, strengths, skill gaps
- 6 Click 'Hear Your Feedback' — TTS plays
- 7 Download PDF report
- 8 Check email — report delivered via n8n

# What Kairix AI Delivers

## AI OUTPUT — PER ANALYSIS

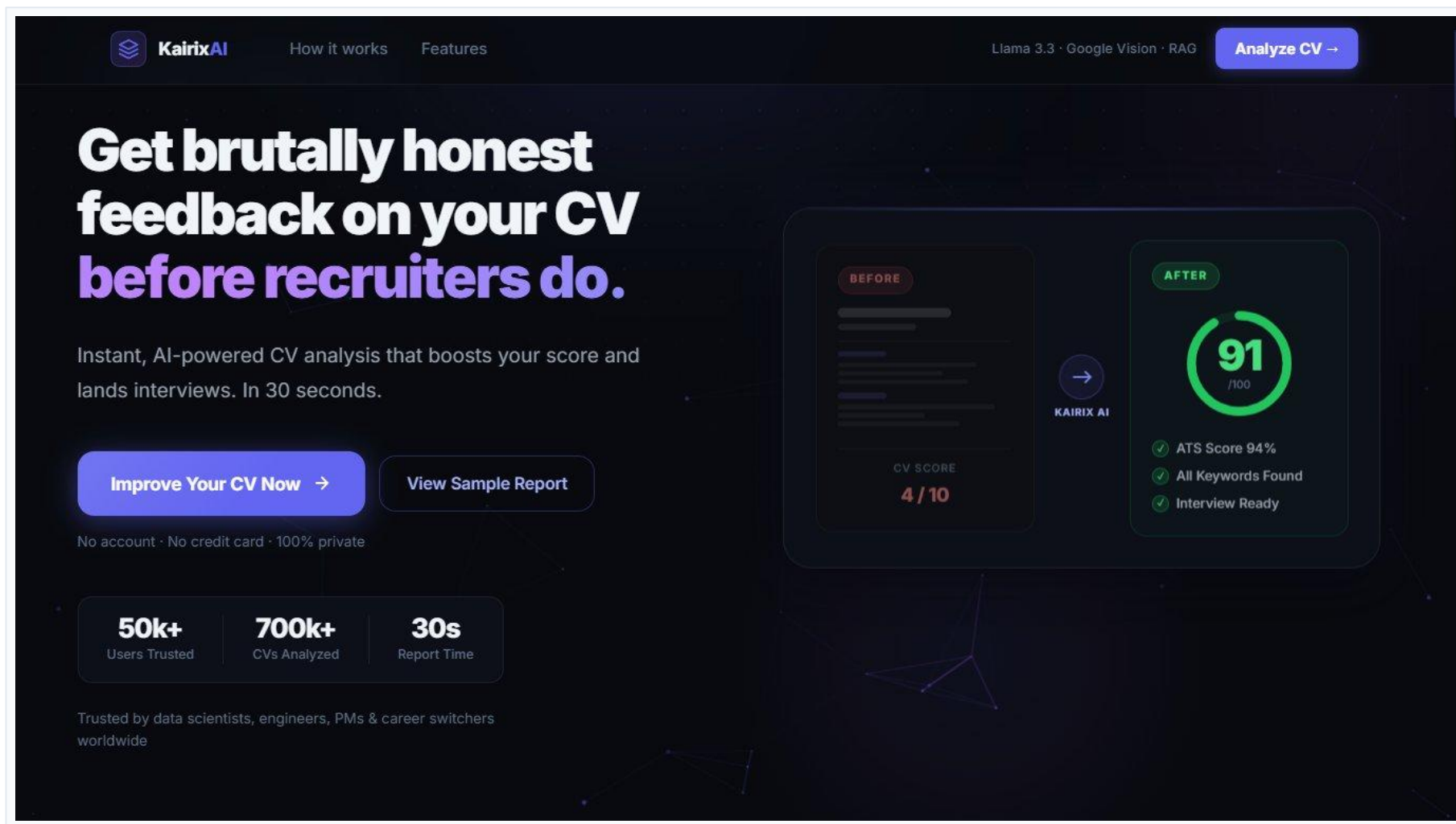
- CV Score out of 100 — visual ring indicator
- Score explanation — why you received that score
- Strengths — what your CV does well
- Skill Gaps — what is missing for the target role
- CV Improvements — specific line-by-line suggestions
- Next Steps — certifications, projects, and actions
- Motivational message personalised to the user
- Text-to-speech playback of the full report
- PDF export of the complete coaching report

## AUTOMATION — EVERY SESSION

- n8n webhook triggered automatically on completion
- Full report emailed via Gmail SMTP — no user action needed
- Session logged to Google Sheets: timestamp, email, score
- Fully event-driven — zero manual intervention

## EVALUATION CRITERIA — COVERED

- ✓ **Innovation** — novel AI career coaching product
- ✓ **Technical integration** — 4 AI systems unified
- ✓ **End-to-end pipeline** — fully functional in production
- ✓ **Real-world problem** — career services domain



## Upload Form — 4-Step Cards

ANALYZE

Get your **Career Intelligence Report**

Free · Private · No account required · Under 30 seconds

1 Upload CV

2 Your Goal

3 Match JD

4 Generate

01 Upload your CV

Accepted: JPG, PNG, PDF, DOCX · Max 10 MB



Drop your CV here, or [browse](#)

PDF · DOCX · JPG · PNG · WEBP

02 Describe your target role

Speak your goal out loud or type it below

 **Click to record your goal**  
Works best in Chrome

"I want to become a senior data analyst at a fintech company in London"

OR TYPE YOUR GOAL

e.g. I want to transition into product management at a B2B SaaS company, ideally from my current software engineering role.

03 Paste a job description

OPTIONAL · UNLOCKS ATS MATCH

Get a keyword match % and ATS analysis against a real job posting

Paste the full job description here — requirements, responsibilities, preferred qualifications...

The AI will simulate an ATS scan and tell you exactly which keywords you're missing.

04 Email your report

OPTIONAL

We'll send the full coaching report to your inbox

you@example.com

No spam. No account required. Unsubscribe anytime.

Generate My Career Report

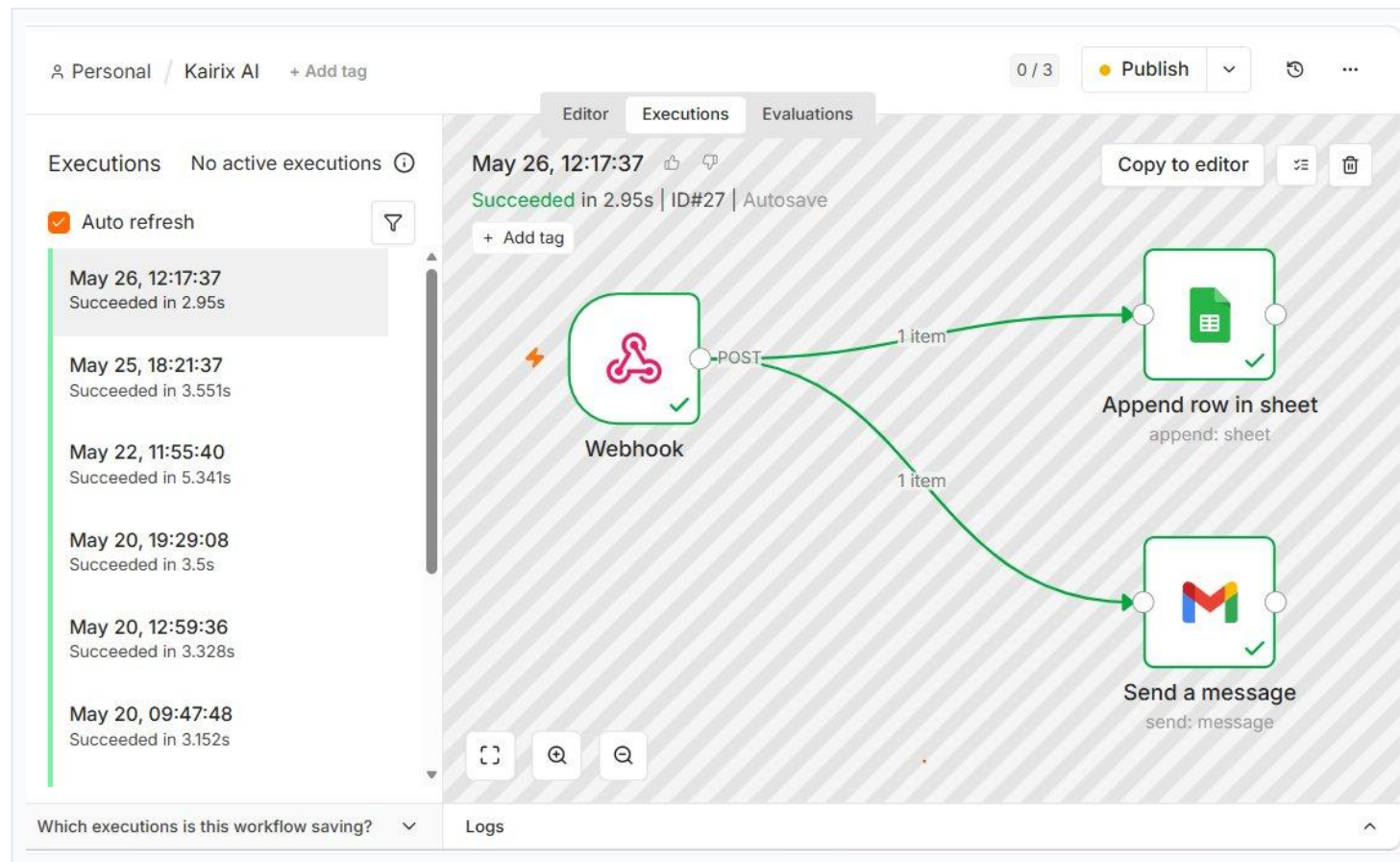
Upload your CV and describe your goal to continue

# Results Screen — Career Intelligence Report

Live analysis output · score ring, JD match, strengths, skill gaps, CV improvements

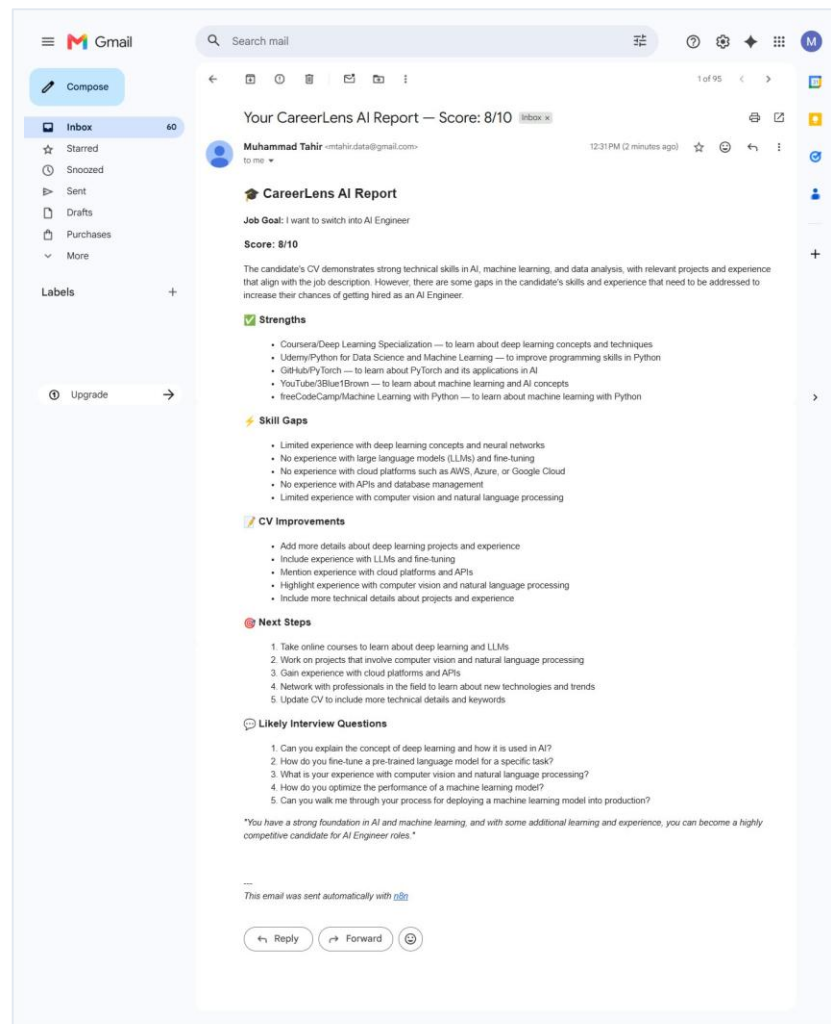


Production workflow on n8n · Webhook trigger splits into Google Sheets logging and Gmail report delivery



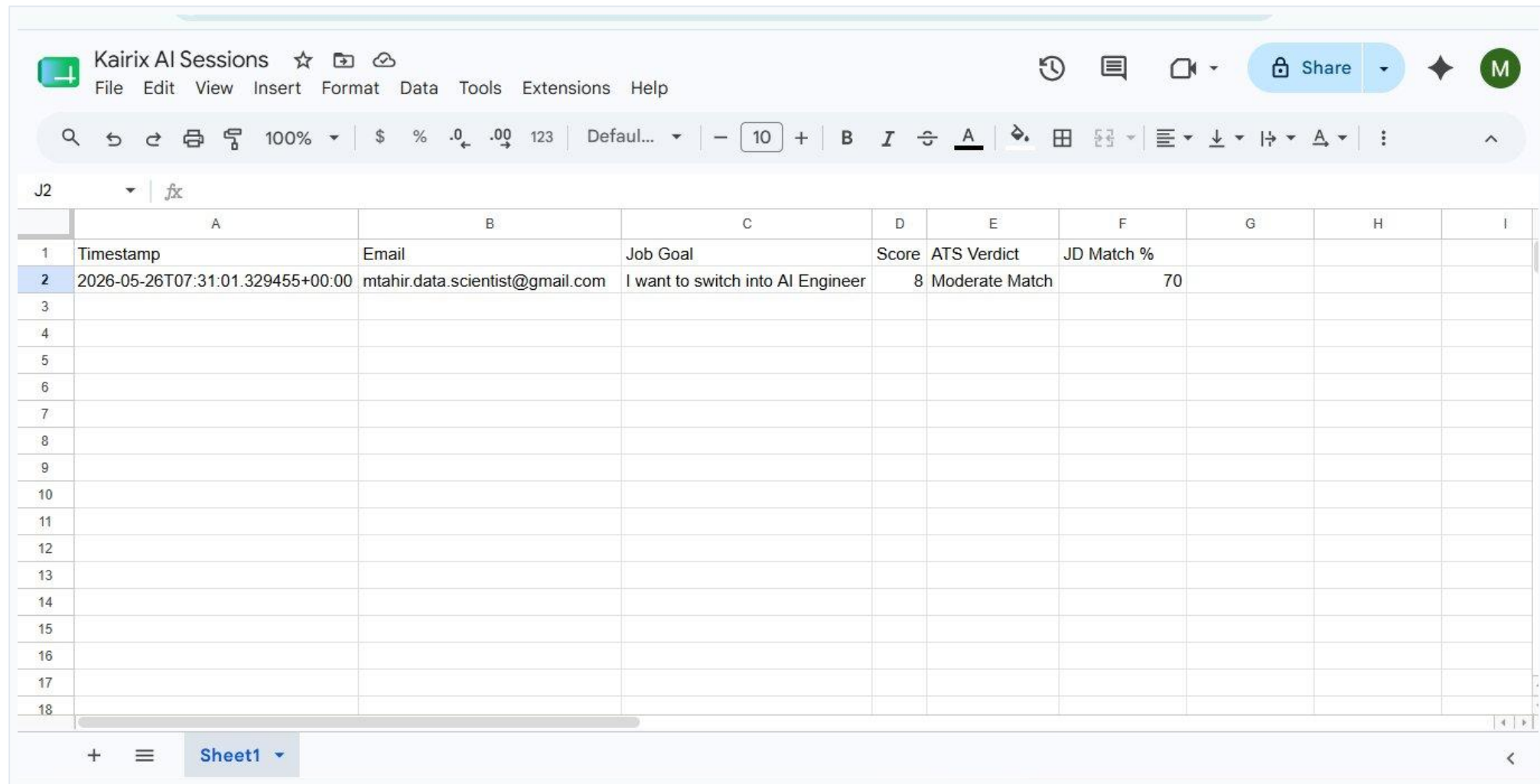
# Email Report — Gmail Inbox

Coaching report delivered automatically by n8n · score, strengths, skill gaps, next steps, interview Qs



# Google Sheets Session Log

Every analysis appended in real time by n8n · timestamp, email, job goal, score, ATS verdict, JD match %



Kairix AI Sessions

File Edit View Insert Format Data Tools Extensions Help

100% 123 Default... 10 B I A

|    | A                                | B                               | C                                 | D     | E              | F          | G | H | I |
|----|----------------------------------|---------------------------------|-----------------------------------|-------|----------------|------------|---|---|---|
| 1  | Timestamp                        | Email                           | Job Goal                          | Score | ATS Verdict    | JD Match % |   |   |   |
| 2  | 2026-05-26T07:31:01.329455+00:00 | mtahir.data.scientist@gmail.com | I want to switch into AI Engineer | 8     | Moderate Match | 70         |   |   |   |
| 3  |                                  |                                 |                                   |       |                |            |   |   |   |
| 4  |                                  |                                 |                                   |       |                |            |   |   |   |
| 5  |                                  |                                 |                                   |       |                |            |   |   |   |
| 6  |                                  |                                 |                                   |       |                |            |   |   |   |
| 7  |                                  |                                 |                                   |       |                |            |   |   |   |
| 8  |                                  |                                 |                                   |       |                |            |   |   |   |
| 9  |                                  |                                 |                                   |       |                |            |   |   |   |
| 10 |                                  |                                 |                                   |       |                |            |   |   |   |
| 11 |                                  |                                 |                                   |       |                |            |   |   |   |
| 12 |                                  |                                 |                                   |       |                |            |   |   |   |
| 13 |                                  |                                 |                                   |       |                |            |   |   |   |
| 14 |                                  |                                 |                                   |       |                |            |   |   |   |
| 15 |                                  |                                 |                                   |       |                |            |   |   |   |
| 16 |                                  |                                 |                                   |       |                |            |   |   |   |
| 17 |                                  |                                 |                                   |       |                |            |   |   |   |
| 18 |                                  |                                 |                                   |       |                |            |   |   |   |

Sheet1

# GitHub Repository

[Code](#) [Issues](#) [Pull requests](#) [Agents](#) [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#) [Settings](#)

 **kairix\_ai** Public Pin Watch 0 Fork 0 Starred 1

main 1 Branch 0 Tags  + Code

 **MuhammadTahir-1-9-4** Fix nav: sticky → fixed so it stays on scroll, frosted glas... ✓ 38b431f · yesterday 19 Commits

|              |                                                               |             |
|--------------|---------------------------------------------------------------|-------------|
| backend      | Add DOCX upload support                                       | 4 days ago  |
| frontend     | Fix nav: sticky → fixed so it stays on scroll, frosted gla... | yesterday   |
| .gitignore   | Initial commit — CareerLens AI full-stack app                 | 2 weeks ago |
| README.md    | Rewrite README — rebrand to Kairix AI, professional...        | 4 days ago  |
| netlify.toml | Add netlify.toml to set publish directory                     | last week   |

 **README**  



### About

No description, website, or topics provided.

-  Readme
-  Activity
-  1 star
-  0 watching
-  0 forks

### Releases

No releases published  
[Create a new release](#)

### Deployments 21